9.2 Infinite Series

An infinite series is the sum of an infinite sequence of numbers

$$a_1 + a_2 + \cdots + a_n + \cdots$$

For example, given the sequence $\left\{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \cdots\right\} = \{a_n\}_{n=1}^{\infty}$

with $a_n = (1/2)^{n-1}$, we can generate the infinite series

$$\sum_{n=1}^{\infty} a_n = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots .$$

Partial Sums

$S_1 = 1 = 2 - 1$

$S_2 = 1 + \frac{1}{2} = \frac{3}{2} = 2 - \frac{1}{2}$

$S_3 = 1 + \frac{1}{2} + \frac{1}{4} = \frac{7}{4} = 2 - \frac{1}{4}$

$S_4 = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{15}{8} = 2 - \frac{1}{8}$

$\vdots$

$S_k = 1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{k-1}} = 2 - \frac{1}{2^{k-1}}$

"The sum of the infinite series $1 + \frac{1}{2} + \frac{1}{4} + \cdots$ is 2."

Definition: Given a sequence $\{a_n\}$, the **infinite series** (or simply the **series**) is an expression of the form

$$a_1 + a_2 + \ldots + a_n + \ldots$$

The number a_n is the n**th term** of the series. The sequence $\{s_n\}$ defined by

$$s_1 = a_1$$
$$s_2 = a_1 + a_2$$
$$\vdots$$
$$s_n = a_1 + a_2 + \ldots + a_n = \sum_{k=1}^{n} a_k$$
$$\vdots$$

is the **sequence of partial sum** of the series, the number s_n being the n**th partial sum**. If the sequence of partial sums converges to a limit L, we say that the series **converges** and that its **sum** is L. In this case, we also write

$$a_1 + a_2 + \ldots + a_n + \ldots = \sum_{n=1}^{\infty} a_n = L.$$

If the sequence of partial sums of the series does not converge, we say that the series **diverges**.

eg. • $1+2+3+4+\cdots$ diverges

• $1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\cdots$ diverges.

Convenient Symbols:

$$\sum_{n=1}^{\infty} a_n \quad \text{or} \quad \sum_{k=1}^{\infty} a_k \quad \text{or simply} \quad \sum a_n$$

Warning: "sequence" and "series"

A series is itself a sequence $\{s_n\}$ obtained from a given sequence $\{a_n\}$ according to the above definition.

Geometric Series

Example: Consider the sequence $\{r^{n-1}\}_{n=1}^{\infty}$ where r is a fixed real numbers, which generates the **geometric series**: $\sum_{n=1}^{\infty} r^{n-1} = 1 + r + r^2 + \ldots + r^{n-1} + \ldots$

If $r = 1$, then $S_n = n \Rightarrow \sum 1^{n-1}$ diverges.

If $r = -1$, then $S_n = \begin{cases} 1 & \text{when } n \text{ is odd} \\ 0 & \text{when } n \text{ is even} \end{cases} \Rightarrow \sum_{n=1}^{\infty} (-1)^{n-1}$ diverges.

For the case when $r \neq \pm 1$.

$$S_n = 1 + r + r^2 + \ldots + r^{n-1}$$

$$rS_n = (r + r^2 + r^3 + \ldots + r^{n-1}) + r^n = S_n - 1 + r^n$$

$$\Rightarrow S_n = \frac{1 - r^n}{1 - r}$$

If $|r| < 1$, $\lim_{n \to \infty} S_n = \frac{1}{1-r}$

If $|r| > 1$, then S_n diverges.

The geometric series $\sum r^{n-1}$ converges to $\frac{1}{1-r}$ if $|r| < 1$, and diverges otherwise.

If $|r| < 1$, the geometric series $a + ar + \ldots + ar^{n-1} + \ldots$ converges to $a/(1 - r)$:

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}, \ |r| < 1.$$

If $|r| \geq 1$, the series diverges.

Example:

1. $a = 1/9$ and $r = 1/3$

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{1}{9} + \frac{1}{9}\cdot\frac{1}{3} + \frac{1}{9}\cdot\left(\frac{1}{3}\right)^2 + \ldots$$

$$= \frac{1/9}{1 - 1/3} = \frac{1}{6}$$

2. $\sum_{n=0}^{\infty} \frac{(-1)^n 5}{4^n}$ $= 5 + 5\left(-\frac{1}{4}\right) + 5\left(-\frac{1}{4}\right)^2 + \ldots$

$a = 5$, $r = -\frac{1}{4}$ $\Rightarrow \sum_{n=0}^{\infty} \frac{(-1)^n 5}{4^n} = \frac{5}{1 - \left(-\frac{1}{4}\right)} = 4$

Further Examples

Example: Consider the series **generated** by the sequence $\{(-1)^n\}_{n=1}^{\infty}$; that is, the series:

Note that $(-1) + (-1)^2 + (-1)^3 + \cdots = -1 + 1 - 1 + 1 - 1 + \cdots$.

$S_n = \begin{cases} -1 & \text{when } n \text{ odd} \\ 0 & \text{when } n \text{ even} \end{cases}$

Geometric series with $a = -1$ & $r = -1$,

so $\sum_{n=1}^{\infty} (-1)^n$ diverges.

Example: The use of "telescoping"

Consider the series

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \frac{1}{3\cdot 4} + \cdots$$

Observe that $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$.

$n=1: \quad \frac{1}{1} - \frac{1}{2}$

$n=2: \quad \frac{1}{2} - \frac{1}{3}$

$n=3: \quad \frac{1}{3} - \frac{1}{4}$

$\vdots$

$n=k: \quad \frac{1}{k} - \frac{1}{k+1}$

$S_k = 1 - \frac{1}{k+1} \Rightarrow \lim_{k\to\infty} S_k = 1$

$\therefore \sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \lim_{k\to\infty} S_k = \lim_{k\to\infty} \left(1 - \frac{1}{k+1}\right) = 1$.

The nth Term Test for a Divergent Series

Example: The series

$$\sum_{n=1}^{\infty} \frac{n+1}{n} = \frac{2}{1} + \frac{3}{2} + \frac{4}{3} + \dots$$

Notice that $\frac{n+1}{n} > 1$ for all $n \geq 1$.

$$> 1 + 1 + 1 + \dots$$

$$\Rightarrow \quad S_k > k \qquad \lim_{k\to\infty} S_k \text{ ``}>\text{''} \lim_{k\to\infty} k \text{ ``}=\text{''} \infty$$

Theorem: If $\sum a_n$ converges, then $a_n \to 0$.

Suppose $\sum a_n$ converges to S.

$$a_n = S_n - S_{n-1}$$

$$\lim_{n\to\infty} a_n = \left(\lim_{n\to\infty} S_n\right) - \left(\lim_{n\to\infty} S_{n-1}\right) = S - S = 0 \quad \Rightarrow \underline{a_n \to 0} \qquad \square.$$

Note that the converse is **not true**!

When $a_n \to 0$, it is not necessarily that $\sum a_n$ converges.

E.g. $\frac{1}{n} \to 0$ but $\sum \frac{1}{n}$ diverges.

The nth Term Test for Divergence

$\sum_{n=1}^{\infty} a_n$ diverges if $\lim_{n\to\infty} a_n$ fails to exist or is different from zero.

Examples of divergent series:

1. $\sum_{n=1}^{\infty} n^2$: $\lim n^2 \to \infty$

2. $\sum_{n=1}^{\infty} \frac{n+1}{n}$: $\lim \frac{n+1}{n} \to 1$

3. $\sum_{n=1}^{\infty} (-1)^{n+1}$ $(-1)^{n+1}$ diverges ($\lim (-1)^{n+1}$ does not exist)

4. $\sum_{n=1}^{\infty} \frac{-n}{2n+5}$ $\lim \frac{-n}{2n+5} = -\frac{1}{2}$

Another divergent series:

The series

$$1 + \underbrace{\left(\frac{1}{2} + \frac{1}{2}\right)}_{=1} + \underbrace{\left(\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}\right)}_{=1} + \ldots + \underbrace{\left(\frac{1}{2^k} + \frac{1}{2^k} + \ldots + \frac{1}{2^k}\right)}_{2^k \text{ times}} + \ldots$$

$> M$ for any M $\Rightarrow$ the series diverges.

Corollary*: Let $\{a_n\}$ be a sequence of nonnegative real numbers. Then the series $\sum a_n$ converges if and only if the sequence $\{s_k\}$ of partial sums is bounded.

Proof ("$\Leftarrow$") If $\{S_k\}$ is a bounded sequence, then $\sum a_n$ converges.

- $\{S_k\}$ is a monotone sequence (increasing sequence: $S_{k+1} \geq S_k$)
- $\{S_k\}$ is bounded (given)

So by the monotone convergence theorem, $\{S_k\}$ converges.

$\Rightarrow$ $\sum a_n$ converges.

Combining Series

Theorem: If $\sum a_n = A$ and $\sum b_n = B$ are convergent series, then

1. Sum Rule: $\sum(a_n + b_n) = \sum a_n + \sum b_n = A + B$

2. Difference Rule: $\sum(a_n - b_n) = \sum a_n - \sum b_n = A - B$

3. Constant Multiple Rule: $\sum ka_n = k\sum a_n = kA$ for any k.

Proof of ①: let $A_n = a_1 + a_2 + \dots + a_n$, $A_n \to A$

$B_n = b_1 + b_2 + \dots + b_n$, $B_n \to B$.

Let S_n be the partial sums of $\sum(a_n + b_n)$.

$S_n = (a_1 + b_1) + (a_2 + b_2) + \dots + (a_n + b_n)$

$= (a_1 + a_2 + \dots + a_n) + (b_1 + b_2 + \dots + b_n) = A_n + B_n$ $\quad \therefore \sum(a_n + b_n)$ converges to $A + B$.

$\lim S_n = \lim(A_n + B_n) = (\lim A_n) + (\lim B_n) = A + B$.

(sum rule for sequences)

Example:

1. $\sum_{n=1}^{\infty} \frac{3^{n-1}-1}{6^{n-1}} = \sum_{n=1}^{\infty} \left(\frac{3^{n-1}}{6^{n-1}} - \frac{1}{6^{n-1}} \right)$

$= \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^{n-1} - \sum_{n=1}^{\infty} \left(\frac{1}{6}\right)^{n-1}$

$= \frac{1}{1-\frac{1}{2}} - \frac{1}{1-\frac{1}{6}} = 2 - \frac{6}{5} = \frac{4}{5}$.

2. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{3}{2^n} = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{3}{2^{n-1} \cdot 2} = \frac{3}{2} \sum_{n=1}^{\infty} \left(\frac{-1}{2}\right)^{n-1}$

$= \frac{3}{2} \frac{1}{1-(-\frac{1}{2})}$

$= 1$

Corollary:

1. Every nonzero constant multiple of a divergent series diverges.
2. If $\sum a_n$ converges and $\sum b_n$ diverges, then $\sum(a_n + b_n)$ and $\sum(a_n - b_n)$ both diverge.

It is not true that if $\sum(a_n + b_n)$ converges, then $\sum a_n$ & $\sum b_n$ converges.

Eg. $a_n = (-1)^n$ & $b_n = (-1)^{n+1}$

$a_n + b_n = 0 \Rightarrow \sum(a_n + b_n)$ converges

but $\sum a_n$ diverges & $\sum b_n$ diverges.

Remarks:

1. Adding or deleting finite terms to or from a series will not affect the series' convergence or divergence. If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=k}^{\infty} a_n$ converges for any natural number k. Conversely, if $\sum_{n=k}^{\infty} a_n$ converges for any $k \geq 1$, then $\sum_{n=1}^{\infty} a_n$ converges.

   Eg: • $\sum_{n=100}^{\infty} \left(\frac{1}{3}\right)^n$ converges

   • $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges $\Rightarrow$ $\sum_{n=10^{1000000}}^{\infty} \frac{1}{n}$ is still a divergent series.

2. Reindexing any series in such a way that the order of its terms is preserved will not alter its convergence. For example, we can write the geometric series $\sum_{n=1}^{\infty} \frac{1}{2^{n-1}} = 1 + \frac{1}{2} + \frac{1}{4} + \ldots$ as

   $\sum_{n=0}^{\infty} \frac{1}{2^n}$, $\sum_{n=5}^{\infty} \frac{1}{2^{n-5}}$ or even $\sum_{n=-4}^{\infty} \frac{1}{2^{n+4}}$.

The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

Proof by contradiction:

Suppose $\sum \frac{1}{n}$ is a convergent series, and it converges to S.

$$S = \left(1 + \frac{1}{2}\right) + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6}\right) + \left(\frac{1}{7} + \frac{1}{8}\right) + \cdots\cdots$$

$$> \left(\frac{1}{2} + \frac{1}{2}\right) + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{6} + \frac{1}{6}\right) + \left(\frac{1}{8} + \frac{1}{8}\right) + \cdots$$

$$= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots\cdots$$

$$= S$$

This contradiction $S > S$ shows that our assumption of convergence must be false. This implies that the harmonic series diverges.

The 2-series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent. [In fact, $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$]

(Apply Corollary*) ← we wish to show that the sequence of partial sums is bounded.

$$S_k = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{k^2}$$

$k_1 = 1 = 2^1 - 1 \Rightarrow S_1 = 1$

$k_2 = 3 = 2^2 - 1 \Rightarrow S_3 = \frac{1}{1} + \left(\frac{1}{2^2} + \frac{1}{3^2}\right) < \frac{1}{1} + \left(\frac{1}{2^2} + \frac{1}{2^2}\right) = 1 + \frac{1}{2}$

$k_3 = 7 = 2^3 - 1 \Rightarrow S_7 = S_3 + \left(\frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \frac{1}{7^2}\right) < 1 + \frac{1}{2} + \left(\frac{1}{4^2} + \frac{1}{4^2} + \frac{1}{4^2} + \frac{1}{4^2}\right)$

$= 1 + \frac{1}{2} + \frac{1}{4}$

$\vdots$

$k_j = 2^j - 1 \Rightarrow S_{k_j} < 1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^{j-1}}$

The RHS are just the partial sums of the geometric series $\sum \frac{1}{2^{n-1}} = 2$.

$\Rightarrow \{S_{k_j}\}$ is bounded by 2. $\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

Extra Examples:

1. Find the nth partial sum of the series

$$1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \ldots + (-1)\frac{1}{2^{n-1}} + \ldots$$

① $S_n = \dfrac{1-(-\frac{1}{2})^n}{1-(-\frac{1}{2})} = \dfrac{1-(-\frac{1}{2})^n}{\frac{3}{2}}$

$S_n \longrightarrow \frac{2}{3}$

2. $\displaystyle\sum_{n=0}^{\infty} \frac{5}{2^n} + \frac{1}{3^n} =$

② $5\left(\sum_{n=0}^{\infty}\left(\frac{1}{2}\right)^n\right) + \left(\sum_{n=0}^{\infty}\left(\frac{1}{3}\right)^n\right)$

$= \dfrac{5}{1-(\frac{1}{2})} + \dfrac{1}{1-\frac{1}{3}}$

$= 10 + \frac{3}{2} = 11\frac{1}{2}$

3. $\sum_{n=1}^{\infty} \frac{40n}{(2n-1)^2(2n+1)^2} =$ $\sum_{n=1}^{\infty}\left[\frac{5}{(2n-1)^2} - \frac{5}{(2n+1)^2}\right]$

4. $\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)} =$

$n=1$: $\frac{5}{1^2} - \frac{5}{3^2}$

$n=2$: $\frac{5}{3^2} - \frac{5}{5^2}$

$n=3$: $\frac{5}{5^2} - \frac{5}{7^2}$

$\vdots$

$n=k$: $\frac{5}{(2k-1)^2} - \frac{5}{(2k+1)^2}$

$\Rightarrow S_k = 5 - \frac{5}{(2k+1)^2}$

$\lim_{k\to\infty} S_k = 5 - \lim_{k\to\infty}\frac{5}{(2k+1)^2} = 5$

④ $\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)} = \sum_{n=1}^{\infty} \left[\frac{1}{2n} - \frac{1}{n+1} + \frac{1}{2(n+2)} \right]$

$= \sum_{n=1}^{\infty} \left(\left[\frac{1}{2n} - \frac{1}{2(n+1)} \right] + \left[\frac{1}{2(n+2)} - \frac{1}{2(n+1)} \right] \right)$

$n=1$: $\left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{6} - \frac{1}{4}\right)$

$n=2$: $\left(\frac{1}{4} - \frac{1}{6}\right) + \left(\frac{1}{8} - \frac{1}{4}\right)$

$n=3$: $\left(\frac{1}{6} - \frac{1}{8}\right) + \left(\frac{1}{10} - \frac{1}{8}\right)$

$\vdots$

$n=k$ $\left(\frac{1}{2k} - \frac{1}{2(k+1)}\right) + \left(\frac{1}{2(k+2)} - \frac{1}{2(k+1)}\right)$

$S_k = \left[\frac{1}{2} - \frac{1}{2(k+1)}\right] + \left[\frac{1}{2(k+2)} - \frac{1}{4}\right]$

$\Rightarrow \lim_{k\to\infty} S_k = \frac{1}{2} - \frac{1}{4} = \underline{\frac{1}{4}}$